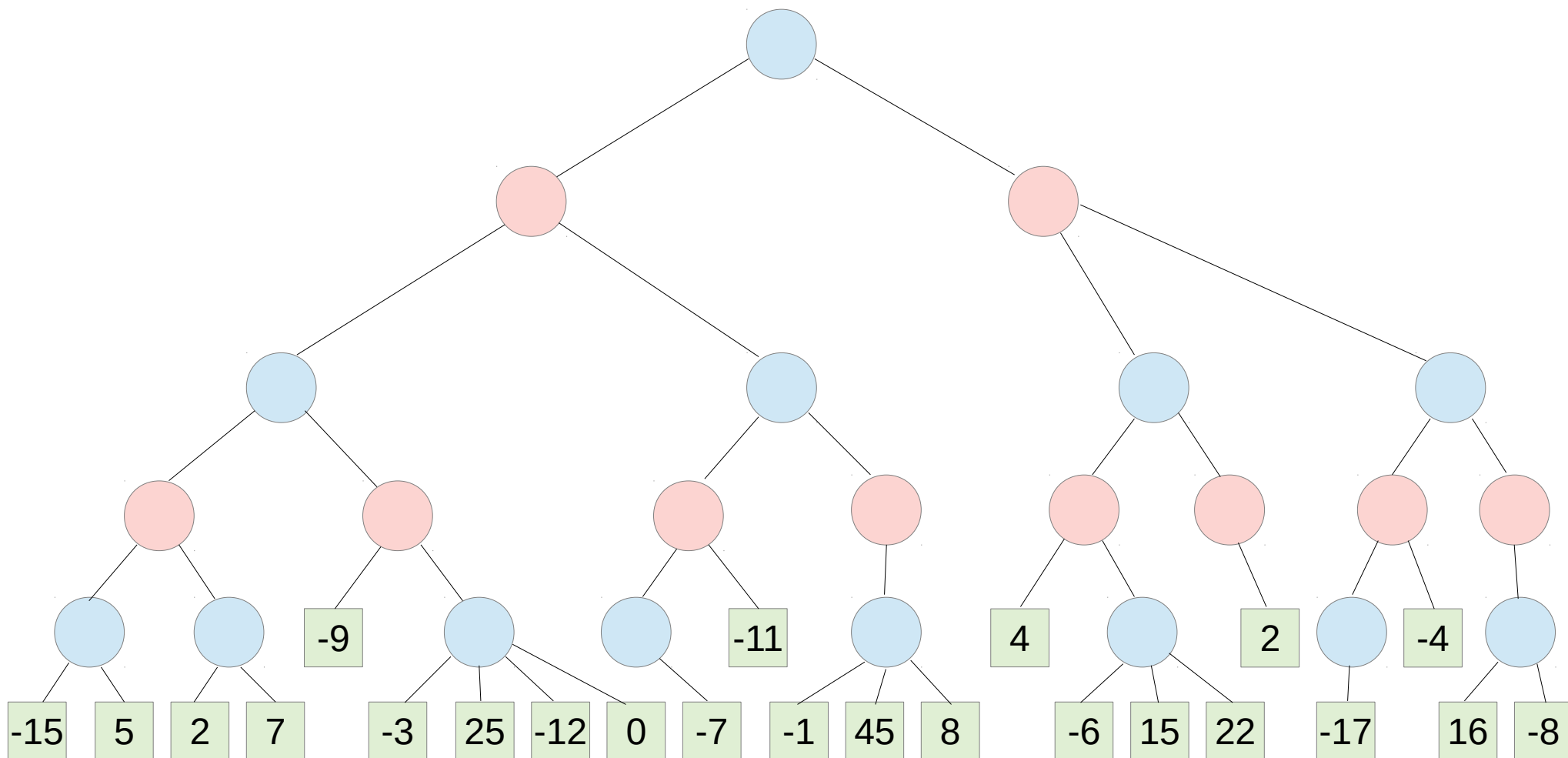
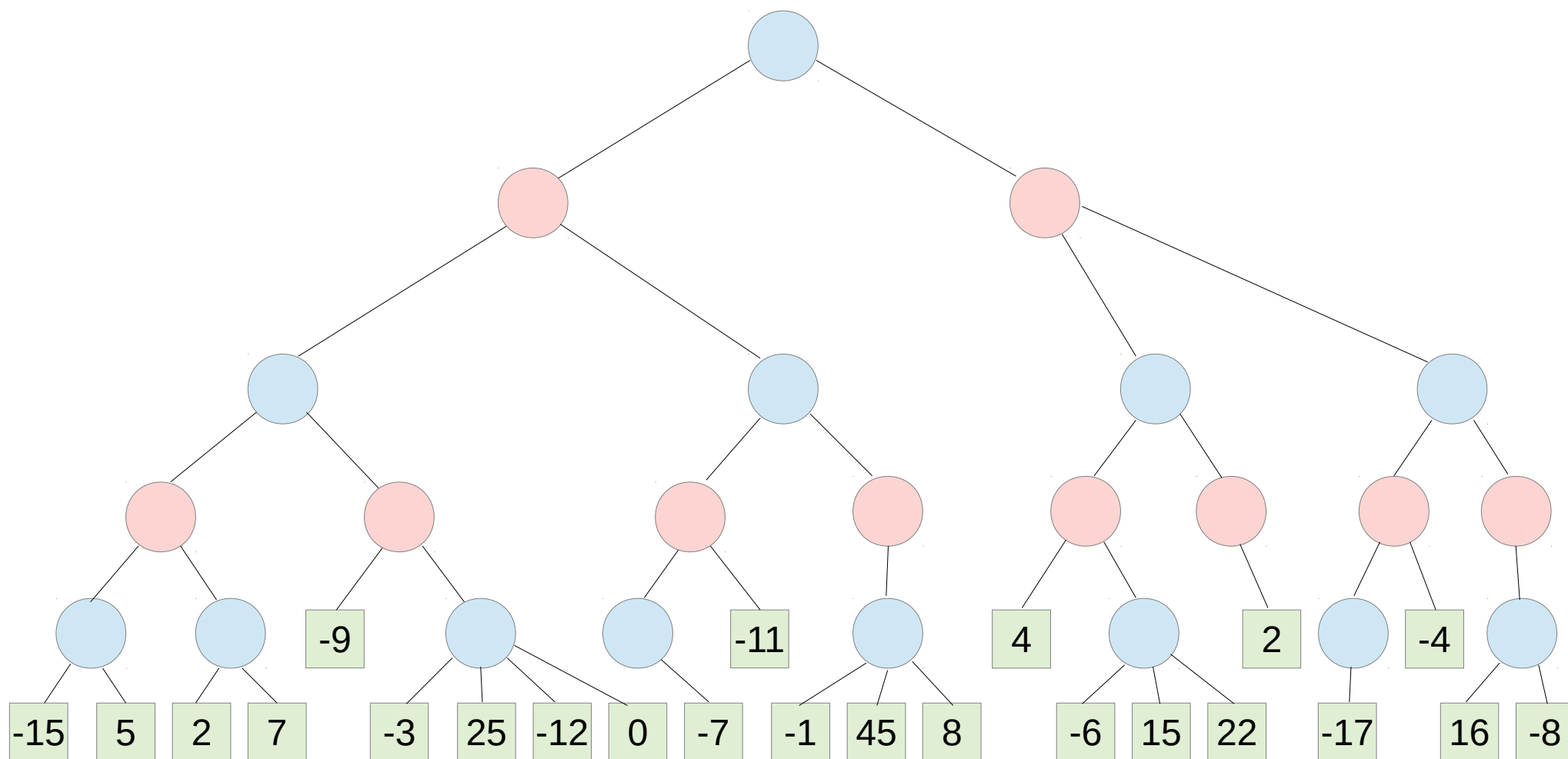




You will be working on the MiniMax tree shown above.

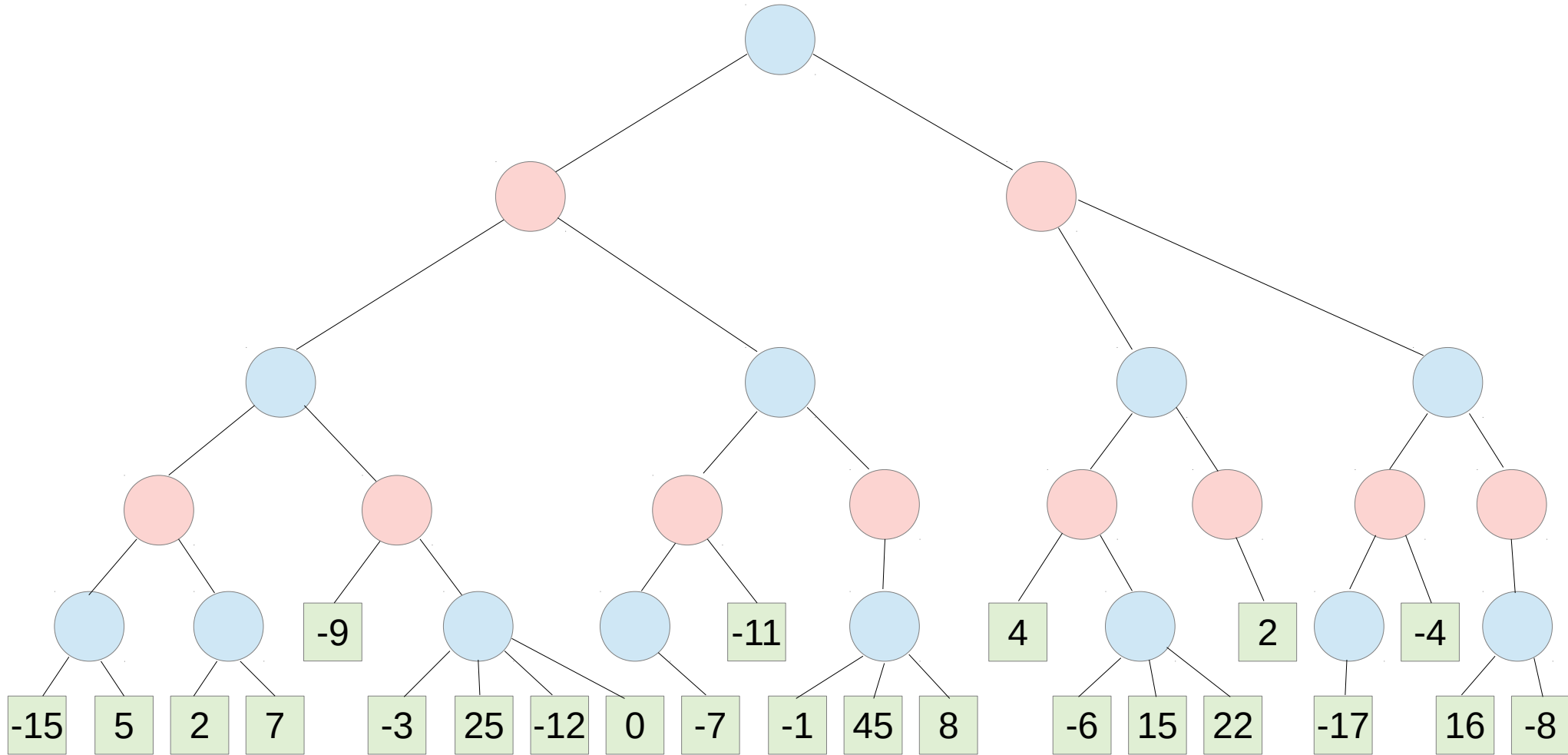
Assume: Player one is **BLUE** Square nodes are terminal nodes.

Remember: MiniMax is a DFS search, ***you must work in correct DFS order or things won't turn out as you would like.*** For nodes at the same level, order of expansion is ***left to right.***



Q1 - On the tree above, write next to each node (including terminals) the **order** in which the node is **expanded** by the MiniMax search.

Q2 - For each level of the tree, indicate whether nodes are **MAX** nodes or **MIN** nodes. How many **complete turns** are there?



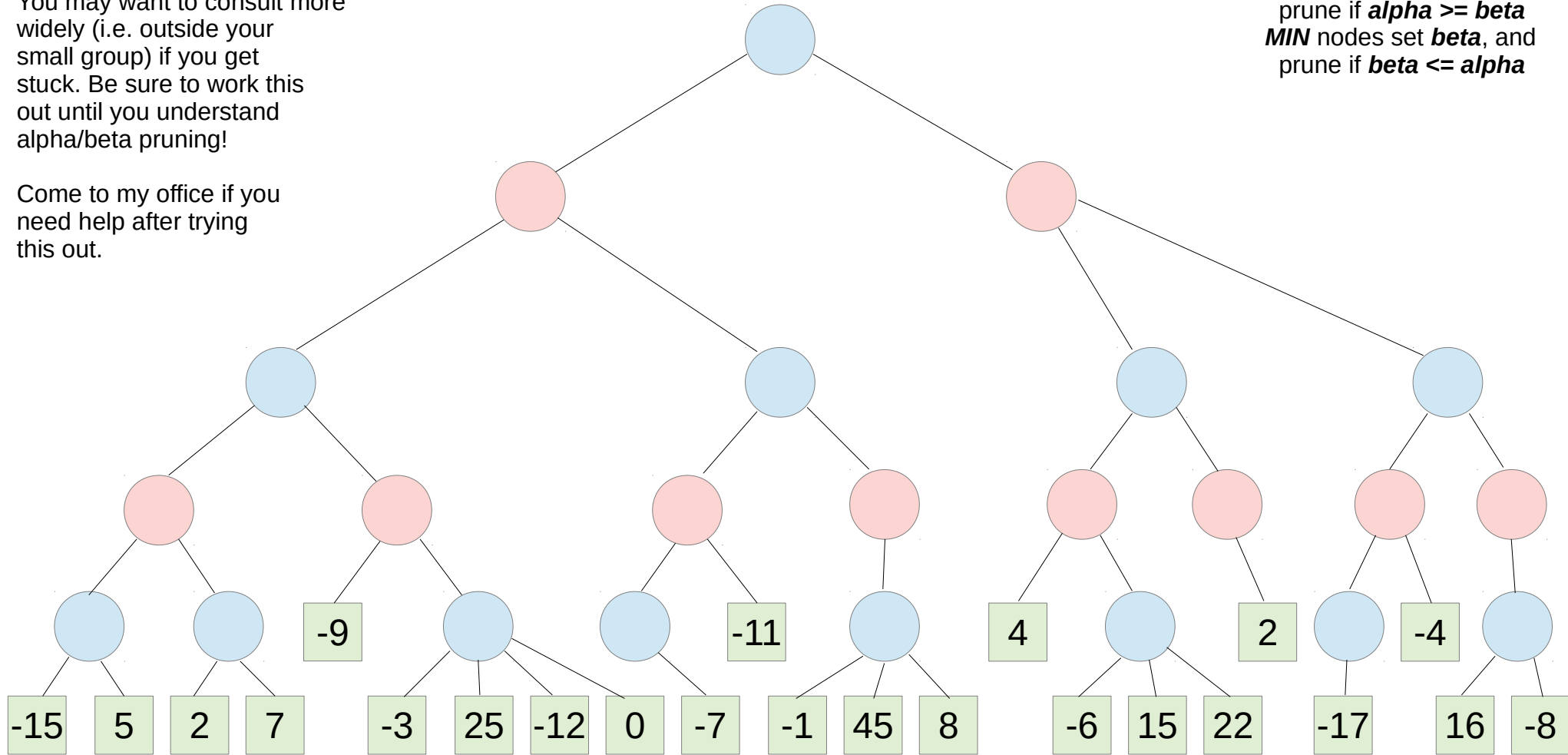
Q3 - Write ***within each node*** the correct ***MiniMax*** score for the node. Do this in the same order in which that score would be computed by the search procedure.

Q4 - Mark the path corresponding to the actions that would be selected by each player all the way to a terminal node. Who wins the game?

You may want to consult more widely (i.e. outside your small group) if you get stuck. Be sure to work this out until you understand alpha/beta pruning!

Come to my office if you need help after trying this out.

Remember:
MAX nodes set **alpha**, and
 prune if **alpha** $\geq$ **beta**
MIN nodes set **beta**, and
 prune if **beta** $\leq$ **alpha**



Q5 – This is the tricky but very important part:

Using the correct **DFS search order**, the **MiniMax scores** for each node, and assuming **alpha=-10000**, **beta=10000** initially for the root node:

Mark which nodes/branches are pruned by **alpha/beta pruning**. And note the alpha or beta value that caused the pruning to take place.

And briefly answer the following question:

How would you use Heuristic Search (A^*) to provide information for your MiniMax utility function? The goal is to think of ways to make the mice play better and avoid being eaten!

Submit this with the rest of the problem set.